DB-Curator

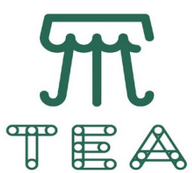
Agentic database construction for scientific evidence

From open-ended scientific questions to verified, auditable database rows

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Application target: build a database, not an answer

Scientific pain point

Specialized databases are still built by slow expert literature curation.

Relevant evidence is scattered across papers, tables, metadata and compound mentions.

A plausible chemical name is not enough; release requires identity, scope and paper provenance.

Agentic framing

The literature set is dynamic, not pre-curated: searches change as candidates and evidence gaps appear.

Each candidate is treated as an evidence contract with mandatory and exclusion predicates.

The loop stops only when the row is supported, rejected, or evidence-limited.

What users get

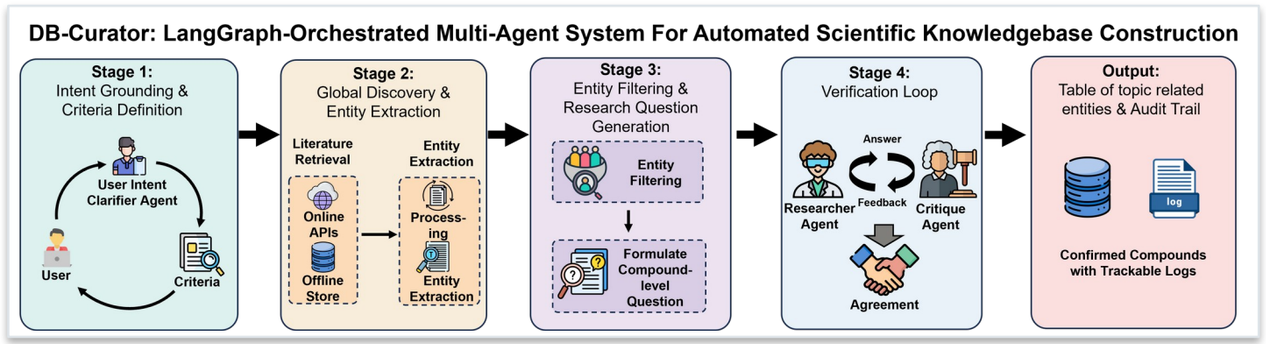
Curated compound rows with identifiers and topic labels.

Cited papers plus quote or local-context support.

Human-reviewable failure states for borderline cases.

ML takeaway: the agent is valuable because it controls retrieval, state tracking and stopping decisions, not because it writes fluent summaries.

How DB-Curator turns intent into verified rows



1. Dynamic literature

Searches are seeded by the user definition, candidate identities and critique-discovered gaps.

2. Candidate grounding

Chemical mentions are normalized, filtered and converted into compound-level verification questions.

3. Predicate verification

Researcher proposes evidence; Critique audits whether each rubric predicate is actually supported.

Figure link: the architecture shows where the corpus evolves and where evidence is audited.

supported **unresolved**

exclusion **exhausted**

Application result: known recovery plus literature-buried additions

246/250

release-accepted rows

Overlap plus human-accepted non-overlap rows.

178

benchmark overlaps

Known database entries recovered automatically.

68

new valid entries

Absent from benchmark, accepted by human experts.

4

errors

False or out-of-scope rows.

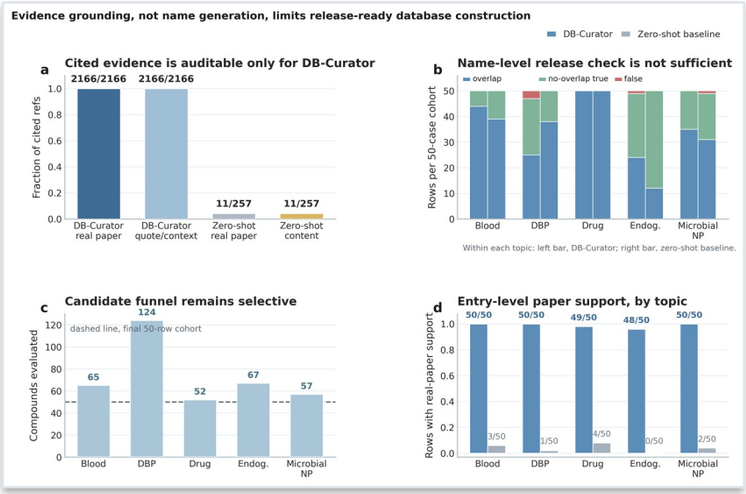


Figure link: panels separate evidence support, name-level release, candidate funnel and entry-level paper support.

Why this matters scientifically

Benchmarks are incomplete for open-ended curation.

A useful system should recover known compounds and surface evidence-backed additions.

DB-Curator shifts human work from full literature search to focused audit of non-overlap rows and evidence.

How we verify the output

Paper-level audit

Resolve the cited item to a real primary paper using DOI, PMID, title or recovered metadata.

Search recovered paper text for the quoted span or equivalent local context.

Exclude compound-database lookup pages from primary-paper citation counts.

Predicate-level audit

Required predicates must be directly supported, not only semantically nearby.

Exclusion contexts such as reagents, solvents or out-of-scope matrices block release.

Unresolved predicates trigger follow-up retrieval or remain evidence-limited.

2,166/2,166

fixed-cohort support

Refs with quote/context support.

7,430/7,453

benchmark-scale support

Auditable refs across five domains.

11/257

zero-shot baseline

Usable refs from direct generation.

Core contrast: baseline models can name plausible compounds, but DB-Curator produces rows whose papers and evidence spans can be audited.

What ML experts should take away

1,887

evaluated candidates

The benchmark-scale test set.

977

system Confirmed

Rows meeting evidence predicates.

49

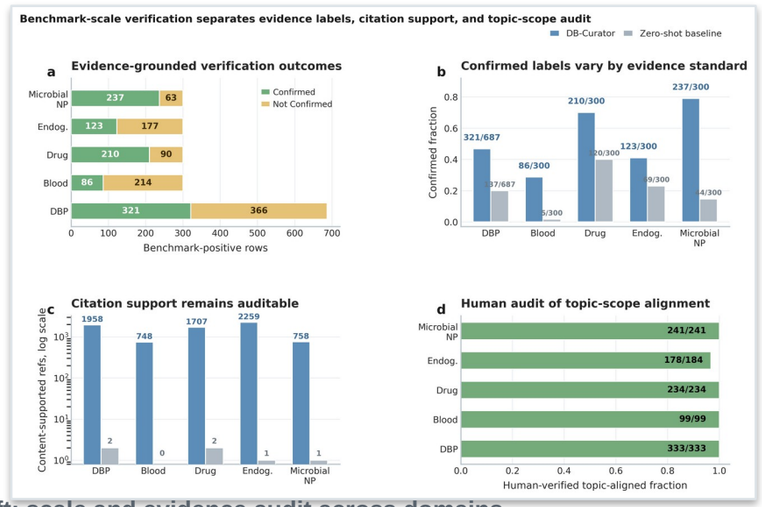
rescued by critique

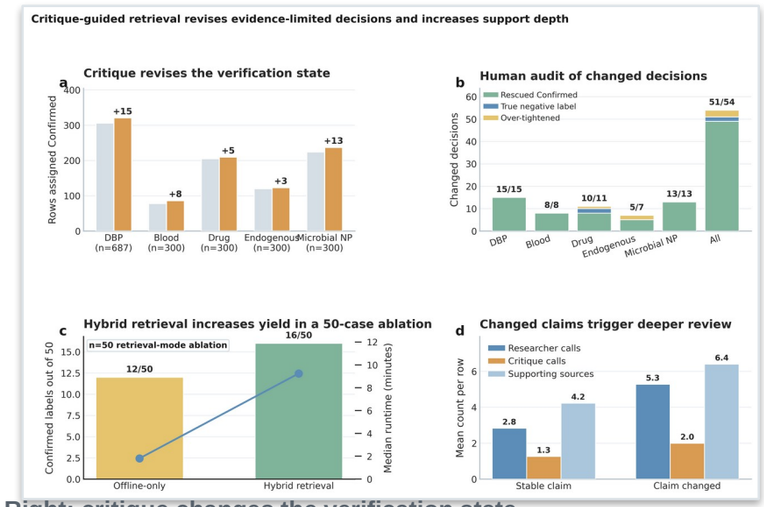
Initially limited rows later confirmed.

5

tightened

Premature Confirmed labels corrected.





Left: scale and evidence audit across domains.

Right: critique changes the verification state.

General lesson: agentic systems are most useful in science when they turn open-ended requests into inspectable states, evidence contracts and auditable artifacts.